Filbin 2018 DIPG — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6 SmartSeq2 samples intermix on one continuous manifold (sample iLISI 1.10→3.40 / 6, plateau θ≥5). One connected component; 0 detached islands. No per-patient archipelago.

Gate 2 — module geography → GO (non-branching)

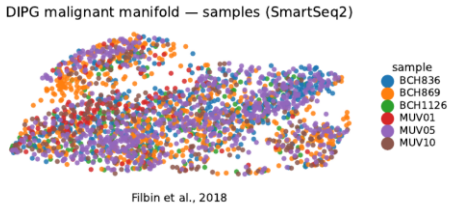
Shared Gast/pSom gradient (smoothed r=+0.64, co-vary) with Crest an inverse terminal readout (r=−0.89 / −0.61). Crest-high decile at pseudotime 0.98; no Gast-arm inversion.

Gate 3 — hub cross-repression → NA

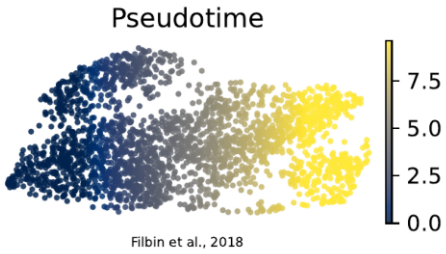
Gast/pSom co-vary at the convergence hub (not compete); both glial arms pSom-dominant, no Gast arm → single collapsed axis, no bifurcation to test (score-hub r is collider-biased).

STEP FIGURES

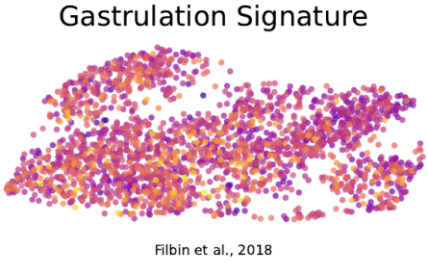
Gate 1: integration



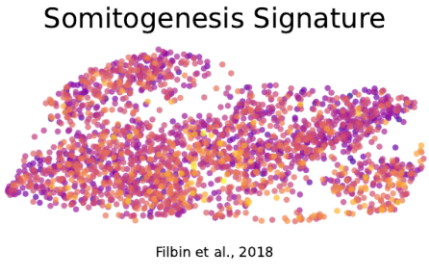
Trajectory / root



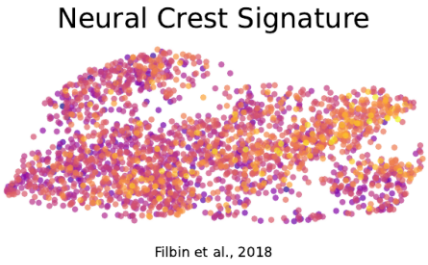
Gast



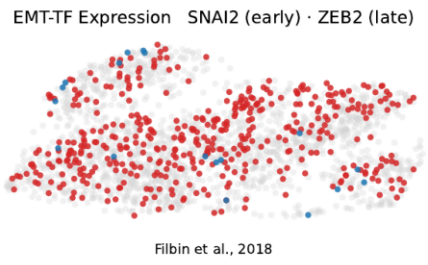
pSom



Crest



EMT-TFs



Dataset. Pediatric H3-K27M mutant glioma (DIPG; Filbin et al., 2018); SmartSeq2 (TPM), 2,259 malignant cells (all 6 samples ≥100; no count gap, max ratio 1.71).

Verdict. ✓ Pattern reproduces as a non-branching embryonic axis: Gast/pSom-high origin → Crest-high terminus; developmental order Gast ρ=−0.77, Crest ρ=+0.86 along pseudotime, checked on a diffusion embedding independent of the fitted UMAP (DC1 ρ=+0.99). Fold-free curve. Geometry: Linear (glial branching architecture, but embryonic modules co-vary).

Parameter choices

Min cells/sample	≥100 (no gap; kept 6/6)	**Integration**	Harmony θ=10; PCA n_pcs=40; spread=1.3
Geometry	Linear / non-branching	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	Gast/pSom-high tip (origin)	**EMT-TFs**	SNAI2 (early) → ZEB2 (late)